

Spatially-aware deep learning for high-resolution snow depth mapping in a Pyrenean catchment: evaluating pattern fidelity with the SPAEF metric

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Abstract

Accurate, high-resolution snow depth (HS) mapping is essential for water-resource management and avalanche forecasting, yet remains challenging due to the strong spatial heterogeneity of the snowpack. We present a deep-learning framework that predicts HS at 1m resolution in the Izas experimental catchment (Central Spanish Pyrenees) from a multi-channel stack of topographic, snow-cover and wind-redistribution predictors derived from a 27-date airborne LiDAR time series (2021–2025). We benchmark a Random Forest (RF) baseline against two convolutional architectures (U-Net and ResUNet++) and show that conventional pixel-wise metrics (e.g. R^2) are insufficient to assess the predicted spatial patterns. We adopt the Spatial Efficiency metric (SPAEF) and a modified, bias-aware variant (MSPAEF) as complementary, pattern-oriented criteria, and introduce a hybrid spatial loss that combines pixel-wise mean squared error with a spatial correlation term weighted by a hyperparameter λ . Across three random seeds, ResUNet++ trained with the spatial loss ($\lambda = 0.4$) achieves the best and most reproducible performance ($R^2 = 0.208 \pm 0.027$, SPAEF = 0.297 ± 0.018), substantially outperforming both the per-pixel Random Forest (SPAEF = 0.107) and a far less stable U-Net ($R^2 = 0.054 \pm 0.124$), whose ranking reverses between R^2 and SPAEF. A leave-one-group-out ablation identifies fine-to-coarse topography (5m DEM derivatives) and snow persistence as the most informative predictors, whereas satellite snow-cover extent contributes

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marginally. Augmenting the input with meteorological forcing (temperature and accumulated precipitation) does not yield a statistically significant improvement, indicating that the topographic and snow-history signals already capture most of the explainable variance. Our results highlight the importance of pattern-aware metrics and losses for high-resolution snow mapping and provide a reproducible benchmark for future work.

Keywords: snow depth, deep learning, ResUNet++, SPAEF, spatial loss, LiDAR, Pyrenees, Random Forest

Highlights

- Deep learning maps 1 m snow depth in a wind-swept Pyrenean catchment.
- A hybrid MSE plus spatial-correlation loss sharpens predicted snow patterns.
- SPAEF reveals spatial errors that R^2 alone overlooks in snow maps.
- ResUNet++ beats a Random Forest and a plain U-Net over three seeds.
- Coarse topography and snow persistence are the key input predictors.

1. Introduction

Seasonal snow is a key component of the mountain water cycle. In Mediterranean mountain ranges such as the Pyrenees, snowmelt sustains river discharge, reservoir storage, hydropower generation and irrigation during the dry season, and the spatial distribution of the snowpack governs avalanche hazard, soil moisture and the timing of vegetation growth [1, 2]. Snow depth (HS) is highly variable in space: wind redistribution, terrain shading, slope, aspect and preferential deposition produce sharp gradients over distances of a few metres, especially in wind-exposed alpine terrain [3, 4]. Capturing this fine-scale variability is therefore essential, but it is precisely where coarse-resolution products and point measurements fail.

Airborne and terrestrial Light Detection and Ranging (LiDAR) surveys have become the reference technique for mapping HS at sub-metre resolution by differencing snow-on and snow-off digital surface models [5, 6]. However, LiDAR acquisitions are expensive and logistically demanding, so they can only be carried out on a limited number of dates per season. This motivates

25 the development of predictive models that can *interpolate and extrapolate* HS from a small set of LiDAR surveys, using spatially exhaustive and inexpensive predictors such as terrain attributes, satellite-derived snow-cover information and meteorological forcing.

Two broad families of methods have been applied to this problem. Statistical and machine-learning regressors (multiple linear regression, generalised additive models, Random Forests, gradient boosting) operate on a per-pixel basis and have proven effective at relating HS to topographic predictors [4, 7]. Their main limitation is that they treat each pixel independently and therefore cannot exploit the spatial context of the snowpack. Convolutional neural networks (CNNs), by contrast, learn spatially structured representations and have recently been used for snow and related cryospheric mapping tasks [8, 9]. Yet most studies still evaluate model quality almost exclusively with pixel-wise metrics – coefficient of determination (R^2), root mean squared error (RMSE), mean absolute error (MAE) – which reward the correct prediction of the mean and penalise outliers, but are largely insensitive to whether the model reproduces the *spatial pattern* of accumulation and scour. A model can attain a respectable R^2 while producing an over-smoothed map that misses the very drift features that matter for avalanche and hydrological applications.

45 In this work we address both the modelling and the evaluation sides of this gap. We build a high-resolution (1 m) HS prediction benchmark for the Izas experimental catchment in the Central Spanish Pyrenees, using a 27-date LiDAR time series (2021–2025) provided in collaboration with the Pyrenean Institute of Ecology (IPE–CSIC). On top of this benchmark we (i) compare a Random Forest baseline with two CNN architectures (U-Net and ResUNet++) under a strict temporal train/validation/test split, (ii) adopt the Spatial Efficiency metric (SPAEF) and a modified, bias-aware variant (MSPAEF) as pattern-oriented evaluation criteria alongside the conventional pixel-wise metrics, and (iii) introduce a hybrid spatial loss that explicitly optimises for spatial pattern fidelity through a tunable correlation term.

1.1. Contributions

The main contributions of this paper are:

1. **A reproducible 1 m snow-depth benchmark** for a Pyrenean catchment, built from airborne LiDAR and a rich stack of 22 topographic,

- 60 snow-cover and wind-redistribution predictors, with a temporally dis-
joint test year (2025) that stresses the extrapolation ability of the mod-
els.
2. **A pattern-aware evaluation protocol** based on SPAEF and MSPAEF,
which we show to be far more discriminative than R^2 for this task –
65 in particular, it exposes a ranking reversal between the Random Forest
and the U-Net (the RF leads on R^2 but the U-Net leads on SPAEF)
and the strong seed-instability of the plain U-Net, both of which a
single-run, R^2 -only assessment would hide.
 3. **A hybrid spatial loss** combining pixel-wise mean squared error with
70 a spatial Pearson-correlation term weighted by a hyperparameter λ ,
together with a complete sweep ($\lambda \in [0, 1]$, eight values, three seeds)
that identifies $\lambda = 0.4$ as the optimum, improving both R^2 and SPAEF
while halving the variance across seeds.
 4. **A leave-one-group-out ablation** of the predictor channels that quan-
75 tifies the relative importance of fine/coarse topography, snow persis-
tence, wind sheltering and satellite snow-cover extent.
 5. **An analysis of meteorological forcing** (air temperature and accu-
mulated precipitation as additional channels), showing that, for this
catchment and predictor set, meteorological inputs do not significantly
80 improve the prediction once topography and snow history are available.

1.2. Paper organisation

The remainder of the paper is organised as follows. Section 2 reviews
related work on snow-depth modelling, deep learning for environmental map-
ping and spatial evaluation metrics. Section 3 describes the study area, the
85 LiDAR-derived dataset and the predictor channels. Section 4 presents the
models, the SPAEF/MSPAEF metrics and the spatial loss, together with
the experimental protocol. Section 5 reports the model comparison, the λ
sweep, the channel ablation and the meteorological experiments. Section 6
discusses the implications and limitations, and Section 7 concludes.

90 2. Related work

2.1. Snow-depth and SWE mapping from terrain and remote sensing

The spatial distribution of mountain snow has traditionally been modelled
by relating snow depth or snow water equivalent (SWE) to terrain attributes

through statistical regression. Elevation, slope, aspect (often decomposed
 95 into northness and eastness), curvature and topographic position indices are
 the classical predictors [10, 11, 4]. Wind redistribution, a dominant control
 in alpine terrain, is commonly represented by terrain-based parameters such
 as the maximum upwind slope S_x [12, 13], which we also adopt here. More
 recently, machine-learning regressors – particularly Random Forests [14] –
 100 have become a de-facto standard for HS/SWE interpolation because they
 handle non-linear interactions and large predictor sets robustly [7, 9, 15].
 These pixel-wise methods are strong baselines but, by construction, ignore
 the spatial arrangement of neighbouring pixels.

2.2. Deep learning for high-resolution environmental mapping

105 Convolutional neural networks have transformed dense prediction tasks
 in remote sensing, from land-cover classification and change detection [8] to
 super-resolution and regression of continuous geophysical fields. Encoder–
 decoder architectures with skip connections, of which the U-Net [16] is the
 archetype, are especially well suited to mapping problems because they com-
 110 bine multi-scale context with spatially precise outputs. Residual and attention-
 augmented variants such as ResUNet [17] and ResUNet++ [18] add residual
 blocks, squeeze-and-excitation and atrous spatial pyramid pooling, which
 stabilise training and improve the modelling of multi-scale structures. In
 the snow context, CNNs and other deep models have been explored for
 115 snow-cover mapping, SWE reconstruction and downscaling [9, 19], but high-
 resolution (metre-scale) HS regression from LiDAR remains comparatively
 under-explored, and systematic comparisons against strong tabular baselines
 under temporally disjoint evaluation are scarce. The work most closely re-
 lated to ours is MAPunet [20], which casts high-resolution HS mapping as
 120 U-Net *pixel-wise regression* from a multi-channel stack of topographic and
 snow-related predictors and applies it over the Swiss Alps (Davos). The
 present study builds directly on that pixel-wise regression framework but (i)
 replaces the plain U-Net backbone with a residual, attention-augmented Re-
 sUNet++; (ii) adds a pattern-oriented spatial loss and pattern-aware metrics
 125 (SPAEF/MSPAEF); and (iii) evaluates under a strictly temporal split with
 a multi-seed stability analysis.

2.3. Evaluation metrics for spatial fields

Pixel-wise metrics (R^2 , RMSE, MAE, bias, Nash–Sutcliffe efficiency) dom-
 inate the snow-modelling literature, yet they assess only the marginal agree-

ment between predicted and observed values and are blind to the spatial organisation of the error. The Spatial Efficiency metric (SPAEF) introduced by (author?) [21] was designed precisely to evaluate the fidelity of spatial patterns, by combining three complementary components: the Pearson correlation between the two fields, the ratio of their coefficients of variation, and the overlap of their value histograms. SPAEF and related multi-component metrics have since been adopted to benchmark distributed hydrological and land-surface models [21, 22]. Despite their suitability, such pattern-oriented metrics are rarely used for high-resolution snow mapping. We argue, and demonstrate empirically, that they are necessary to obtain an honest picture of model quality and stability, and we further use them – through a differentiable spatial-correlation term – to *guide* the optimisation, not merely to evaluate it.

2.4. Pattern-oriented and physics-aware losses

Optimising solely a pixel-wise loss (MSE/MAE) tends to produce regression to the mean and over-smoothed fields. To counteract this, a growing body of work augments the training objective with structural or perceptual terms – for example structural similarity (SSIM) losses, gradient-difference losses, or adversarial terms – so that the model is rewarded for reproducing spatial structure [23]. In the environmental domain, correlation-based and multi-scale loss terms have been used to better preserve spatial texture. Building on this idea, we propose a simple and interpretable hybrid loss that linearly blends MSE with a spatial Pearson-correlation penalty controlled by a single hyperparameter λ , and we characterise its effect on both accuracy and spatial pattern fidelity through a full sweep.

3. Study area and data

3.1. Study area

The study is carried out in the Izas experimental catchment, a small ($\sim 0.55 \text{ km}^2$) alpine headwater basin located in the Central Spanish Pyrenees (province of Huesca), at elevations of approximately 2000–2300 m a.s.l. The catchment has been monitored for decades by the Pyrenean Institute of Ecology (IPE–CSIC) and is characterised by a strongly wind-influenced snowpack: persistent north-westerly winds generate pronounced snow redistribution, with deep drifts on lee slopes and scoured wind-exposed ridges. This produces the sharp, repeatable spatial patterns of accumulation that

165 make Izas an ideal testbed for high-resolution HS prediction (Fig. 1). The LiDAR and meteorological data used in this study were provided through a collaboration with IPE-CSIC.

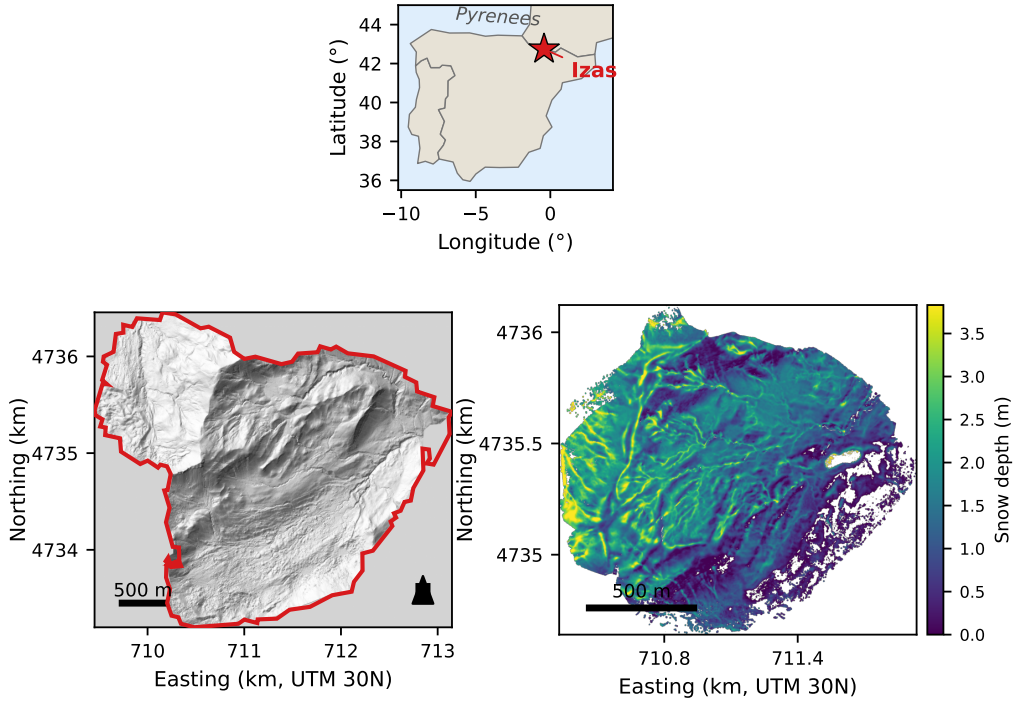


Figure 1: Location and physiography of the Izas experimental catchment (Central Spanish Pyrenees). (a) Regional setting within the Iberian Peninsula and the Pyrenees; (b) outline of the LiDAR-surveyed footprint (union of all acquisition dates) over a hillshade of the 1 m DEM (EPSG:25830, UTM zone 30N); (c) example LiDAR-derived snow-depth map (1 April 2025) illustrating the wind-driven accumulation patterns the models must reproduce.

3.2. LiDAR snow-depth surveys

170 Snow depth was derived from airborne LiDAR surveys by differencing snow-on digital surface models against a snow-off reference, yielding HS maps at 1 m spatial resolution. A total of 27 acquisition dates spanning four hydrological years (2021–2025) are available. The resulting HS rasters constitute the prediction target (ground truth) of all models. To enable convolutional processing, each dated HS map and its co-registered predictor stack are tiled

175 into fixed-size patches; pixels outside the catchment mask or without valid
HS are excluded from both training and evaluation.

3.3. Predictor channels

Each training sample is a multi-channel image in which the predictors
are spatially co-registered with the target HS map at 1 m resolution. The
180 base configuration comprises **22 channels**, which we group into four families
(Table 1); Fig. 2 shows an example predictor stack for one tile:

- **Fine-scale topography (1 m).** Elevation (DEM), slope, northness
(cos of aspect) and eastness (sin of aspect), and the Topographic Posi-
tion Index (TPI). These describe the local terrain that controls radia-
185 tion, gravitational transport and small-scale deposition.
- **Coarse-scale topography (5 m).** The same five terrain derivatives
computed from a 5 m DEM. The coarser scale captures the meso-
topographic context (ridges, gullies, broad lee slopes) that governs wind
redistribution over longer fetch distances.
- 190 • **Wind sheltering.** The maximum upwind slope index S_x [12] com-
puted over a 200 m search distance for eight azimuths (0° , 45° , $\dots$,
 315°). These eight channels encode directional exposure/sheltering and
are the primary terrain-based proxy for wind-driven accumulation and
scour.
- 195 • **Snow-cover history.** The fractional snow-cover persistence over the
15, 30 and 60 days preceding each acquisition, derived from a binary
satellite snow-cover product (snow vs. no-snow) reprojected to 1 m and
temporally averaged; plus the instantaneous satellite snow-cover extent
(SCE) of the acquisition date.

200 *Snow persistence.* The persistence channels summarise the recent snow his-
tory of each pixel. For a given acquisition date and temporal window $W \in$
 $\{15, 30, 60\}$ days, the binary snow-cover product (snow = 1, no-snow/cloud
= 0) is reprojected from its native 10 m resolution to 1 m by bilinear re-
sampling and averaged over all available observations within the window,
205 yielding a value in $[0, 1]$ that represents the fraction of recent days on which
the pixel was snow-covered. The ablation study (Section 5.3) shows that
these channels are among the most informative predictors.

Table 1: The 22 base predictor channels and the four additional meteorological channels (26-channel configuration). “Static” channels are constant in time; “dynamic” channels vary between acquisition dates.

Idx	Channel	Family	Type
0	DEM (1 m)	Topo 1 m	static
1	Slope (1 m)	Topo 1 m	static
2	Northness (1 m)	Topo 1 m	static
3	Eastness (1 m)	Topo 1 m	static
4	TPI (1 m)	Topo 1 m	static
5	SCE (snow-cover extent)	Snow	dynamic
6–13	S_x 200 m (8 azimuths)	Wind	static
14	Persistence 15 d	Snow	dynamic
15	Persistence 30 d	Snow	dynamic
16	Persistence 60 d	Snow	dynamic
17	DEM (5 m)	Topo 5 m	static
18	Slope (5 m)	Topo 5 m	static
19	Northness (5 m)	Topo 5 m	static
20	Eastness (5 m)	Topo 5 m	static
21	TPI (5 m)	Topo 5 m	static
<i>Additional meteorological channels (26-ch configuration)</i>			
22	Mean air temp. 7 d	Meteo	dynamic (scalar)
23	Mean air temp. 15 d	Meteo	dynamic (scalar)
24	Mean air temp. 30 d	Meteo	dynamic (scalar)
25	Accum. precip. from Oct	Meteo	dynamic (scalar)

Meteorological channels (extended configuration).. To test whether explicit meteorological forcing improves prediction, we build an extended 26-channel configuration that adds four scalar channels derived from the catchment automatic weather station (provided by IPE–CSIC): the mean air temperature over the 7, 15 and 30 days preceding the acquisition, and the precipitation accumulated since the 1st of October of the corresponding hydrological year. These are scalar (spatially uniform) per date: every pixel of a given acquisition shares the same value, so the channels encode the temporal meteorological context rather than spatial structure.

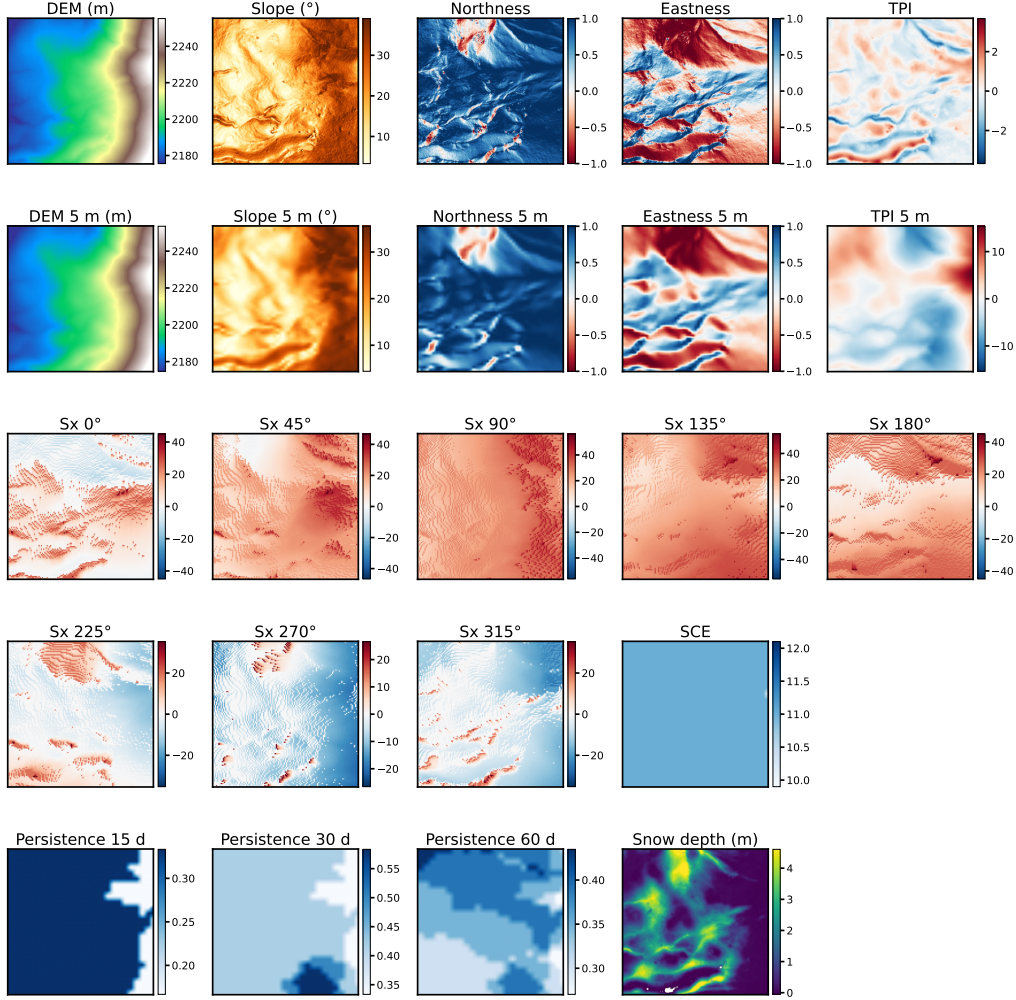


Figure 2: Example predictor stack for a single 256×256 m tile (2 February 2021). Rows, top to bottom: fine 1 m topographic derivatives (DEM, slope, northness, eastness, TPI); the corresponding coarse 5 m derivatives; the eight S_x wind-sheltering indices ($0-315^{\circ}$); and the satellite snow-cover extent (SCE), the 15/30/60-day snow-persistence channels and the target LiDAR snow-depth map. Each panel is independently scaled; diverging fields are centred at zero.

3.4. Temporal split

To obtain an honest estimate of predictive skill on *unseen conditions*, we use a strictly temporal train/validation/test split rather than a random pixel or tile split, which would leak spatial information between sets (Fig. 3):

- **Training:** hydrological years 2021–2023 (15 dates, 4241 tiles).
- **Validation:** 2024 (7 dates, 431 tiles).
- **Test:** 2025 (5 dates, 220 tiles).

The test year 2025 was an exceptionally snowy season: the mean test HS (1.09 m) is 51% higher than the mean training HS (0.72 m). This distribution shift is deliberate – it stresses the extrapolation ability of the models and explains the moderate absolute R^2 values and the non-negligible seed-to-seed variance observed in Section 5. We regard robustness under this shift as a key, realistic requirement for operational snow mapping.

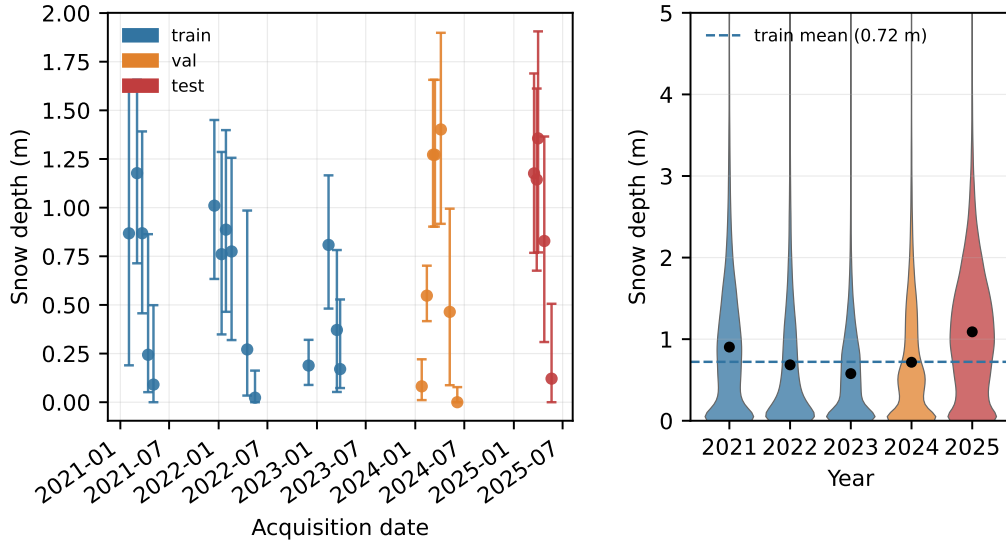


Figure 3: Temporal train/validation/test split and the corresponding distributions of LiDAR snow depth. (a) Per-date median snow depth ($\pm$ interquartile range) along the acquisition timeline, coloured by split. (b) Per-year snow-depth distributions (violins); black dots mark the annual means and the dashed line the training mean. The 2025 test year is markedly snowier than the 2021–2023 training years, illustrating the distribution shift that the models must extrapolate to.

230 4. Methods

4.1. Overview and preprocessing

All models predict HS at 1 m resolution from the predictor stack described in Section 3.3. Each channel is normalised to a comparable range using fixed, physically motivated scalings (e.g. elevation is standardised with a reference
 235 mean and standard deviation, slopes are divided by 90° , the TPI is clipped to $[-1, 1]$, the S_x indices are divided by 90° , persistence channels are already in $[0, 1]$, and SCE is binarised). Invalid sentinel values (-9999) and NaNs are replaced by zero after normalisation. Pixels without valid HS are masked out of the loss and of every metric. Crucially, normalisation is applied to
 240 the full 22-channel stack *before* any channel subset is selected, so that the ablation experiments (Section 5.3) do not alter the normalisation statistics of the remaining channels.

4.2. Models

Random Forest (baseline).. As a strong non-spatial baseline we train a Ran-
 245 dom Forest regressor [14] on the per-pixel feature vectors. Hyperparameters were optimised with Optuna [24] and the final configuration uses 300 trees, a maximum depth of 15, a minimum of one sample per leaf, $\log_2$ feature subsampling and a minimum of two samples to split. The RF is trained on the union of the training and validation pixels (standard refit after hyperpa-
 250 rameter optimisation) and predicts each pixel independently; spatial metrics are computed by reassembling the per-pixel predictions into their original tiles.

U-Net.. The U-Net [16] is a symmetric encoder–decoder with skip connec-
 255 tions. Our implementation uses a base width of 48 feature maps and the standard four-level contracting/expanding path, and is trained end-to-end as a pixel-wise regressor. We use Group Normalisation in the U-Net as well, since Batch Normalisation proved unreliable under our small batch sizes – its running statistics collapsed at evaluation time – and using the same normal-
 260 isation in both networks ensures that the U-Net/ResUNet++ comparison isolates the effect of the architecture rather than of the normalisation layer.

ResUNet++.. ResUNet++ [18] augments the residual U-Net [17] with squeeze-
 and-excitation blocks in the encoder, an atrous spatial pyramid pooling (ASPP) bridge, and attention gates in the decoder. These components im-
 prove gradient flow and multi-scale context aggregation. We use a base width

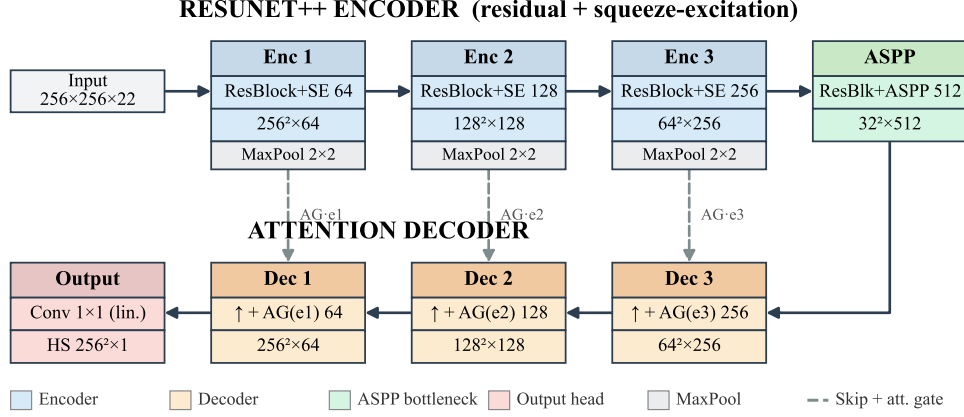


Figure 4: ResUNet++ adapted for pixel-wise snow-depth regression. A three-level residual encoder with squeeze-and-excitation (SE) blocks and Group Normalisation encodes the N -channel input; an atrous spatial pyramid pooling (ASPP, dilations $r = 1, 6, 12, 18$) bottleneck aggregates multi-scale context; and a decoder with attention-gated skip connections (AG) reconstructs the field. The 1×1 output head is linear (no activation), yielding a continuous snow-depth map. Resolutions and channel counts correspond to the base-width-64 configuration on 256×256 tiles.

265 of 64 and Group Normalisation (8 groups) instead of Batch Normalisation, which we found essential for stable training under the small batch sizes imposed by high-resolution tiles and the heterogeneous, date-dependent statistics of the snowpack.

4.3. Adapting ResUNet++ for pixel-wise snow-depth regression

270 ResUNet++ was originally proposed for binary medical image *segmentation* [18]; running it as a snow-depth predictor required adapting it to the pixel-wise regression framework introduced by MAPunet [20]. Concretely, the model learns a mapping $f : \mathbb{R}^{H \times W \times N} \rightarrow \mathbb{R}_{\geq 0}^{H \times W}$ that turns an N -channel input volume (here $N = 22$, or 26 with the meteorological channels) into a single continuous snow-depth field (Fig. 4). We made the following modifications:

275

- **Input layer.** The first convolution was widened to accept the N co-registered predictor channels (Table 1) instead of a fixed three-band image, so that arbitrary channel subsets can be fed to the network (used by the ablation in Section 5.3).

280

- **Regression output.** The final 1×1 convolution produces a single feature map and, crucially, carries *no* output activation (no sigmoid/softmax), yielding unbounded continuous values; predictions are clipped to non-negative depths only at inference.
- 285 • **Masked regression loss.** High-resolution LiDAR maps contain *no-data* pixels (occlusions, gaps, out-of-catchment cells). Following the masked-supervision principle of MAPunet, the loss and all metrics are evaluated only over valid pixels; invalid pixels are excluded from both the pixel-wise MSE and the spatial-correlation term of Eq. (4).
- 290 • **Normalisation for heterogeneous inputs.** Group Normalisation (8 groups) replaces Batch Normalisation to remain stable under the small batches imposed by high-resolution tiles and the strongly date-dependent statistics of the snowpack, and every channel is scaled with fixed physical factors (Section 4.1) before being fed to the network.

295 The same adaptations apply to the U-Net baseline, so that the two architectures differ only in their internal blocks and are compared on an equal footing.

4.4. Spatial evaluation metrics

300 *Pixel-wise metrics.* We report the coefficient of determination (R^2 , equivalently the Nash–Sutcliffe efficiency), the mean absolute error (MAE), the root mean squared error (RMSE) and the bias, all computed over the valid test pixels.

SPAEF. To assess the fidelity of the predicted *spatial pattern* we use the Spatial Efficiency metric [21], defined as

$$\text{SPAEF} = 1 - \sqrt{(\alpha - 1)^2 + (\beta - 1)^2 + (\gamma - 1)^2}, \quad (1)$$

305 where the three components measure complementary aspects of pattern agreement:

$$\alpha = \rho(o, s), \quad \beta = \frac{\sigma_s/\mu_s}{\sigma_o/\mu_o}, \quad \gamma = \frac{\sum_{j=1}^K \min(h_j^o, h_j^s)}{\sum_{j=1}^K h_j^o}. \quad (2)$$

Here o and s are the observed and simulated fields, ρ is the Pearson correlation coefficient (α measures linear co-location of the pattern), β is the ratio of the coefficients of variation (it penalises over-/under-dispersed patterns), and

310 γ is the intersection of the K -bin histograms h^o and h^s of the two fields (it measures the agreement of the value distributions). A perfect match yields SPAEF = 1; the metric is unbounded below. SPAEF is computed per tile and averaged over the test set.

MSPAEF.. As a second pattern-oriented metric we use the modified SPAEF
 315 (MSPAEF) of (author?) [25],

$$\text{MSPAEF} = 1 - \frac{1}{4}[(\alpha - 1)^2 + \beta^2 + \gamma^2 + \delta^2], \quad (3)$$

where $\alpha = \rho(o, s)$ is again the Pearson correlation, β is the RMSE normalised by the interquartile range (IQR) of the observations, γ is the absolute mean bias normalised by the same IQR, and δ is a standard-deviation-ratio term. Compared with the classical SPAEF of Eq. (1), MSPAEF (i) requires no
 320 histogram-bin parameter and (ii) is explicitly sensitive to the absolute mean bias, which SPAEF ignores. This is particularly relevant here, since all models systematically under-predict the anomalously snowy 2025 test year; we therefore report MSPAEF as a complementary, bias-aware indicator. Like SPAEF, it is computed per tile and averaged over the test set.

325 4.5. Hybrid spatial loss

Training a CNN with a pure pixel-wise MSE encourages regression to the mean and tends to produce over-smoothed maps. To explicitly reward spatial pattern fidelity, we train the CNNs with a hybrid loss that linearly combines the pixel-wise MSE with a spatial Pearson-correlation term:

$$\mathcal{L} = (1 - \lambda) \underbrace{\text{MSE}(o, s)}_{\text{pixel-wise}} + \lambda \underbrace{(1 - \rho_{\text{spatial}}(o, s))}_{\text{pattern}}, \quad (4)$$

330 where ρ_{spatial} is the Pearson correlation between the predicted and observed fields computed over the valid pixels of each tile, and $\lambda \in [0, 1]$ controls the trade-off between absolute accuracy and pattern agreement. With $\lambda = 0$ the loss reduces to plain MSE; increasing λ progressively prioritises the reproduction of the spatial pattern. The correlation term is differentiable and
 335 adds negligible computational cost. We characterise the effect of λ through a full sweep (Section 5.2).

4.6. Training and evaluation protocol

Implementation and hardware.. All models were implemented in PyTorch 2.11 (CUDA 12.8) and trained on a single desktop workstation with an NVIDIA GeForce RTX 5060 Ti GPU (16 GB), an AMD Ryzen 5 3400G CPU and 32 GB of RAM, running Windows 10. The Random Forest used scikit-learn 1.8, and all hyperparameter searches used Optuna 4.8 [24].

Hyperparameter optimisation.. All model hyperparameters were tuned with Optuna [24] using a Tree-structured Parzen Estimator (TPE) sampler on the train/validation split, keeping the 2025 test year strictly untouched during the search. For the CNNs we jointly searched the base width, optimiser, learning rate, weight decay, dropout, batch size and gradient clipping (Table 2); each trial was trained for up to 150 epochs with a median pruner (10 start-up trials, 20 warm-up epochs) and scored by the validation MSE. For the Random Forest we searched the number of trees, the maximum depth, the minimum number of samples per leaf and per split, and the feature-subsampling rule, scoring trials by the validation R^2 . The best configuration of each model (Table 2) was then fixed for all subsequent experiments; the spatial-loss weight λ was tuned separately (Section 5.2) on top of the chosen ResUNet++ backbone, and the final models were retrained for 50 epochs (see below).

Training.. All CNN models are trained for 50 epochs without early stopping and without data augmentation, to keep the comparison controlled and reproducible. Every model is evaluated at the checkpoint with the lowest validation loss (best-validation epoch), applying the same selection rule to all architectures for a fair comparison.

Reproducibility and seeds.. Every CNN experiment is repeated with three random seeds (42, 123, 7), with the seed fixed immediately before model construction. We report the mean and standard deviation across seeds for all metrics; the seed spread is an essential part of our results, as it reveals stability differences that single-run numbers would mask. The RF, being almost deterministic for this problem, is likewise evaluated over the three seeds for consistency.

Experiments.. Four experiments are reported in Section 5: (i) a head-to-head comparison of RF, U-Net and ResUNet++ on the 22-channel dataset; (ii) a sweep of the spatial-loss weight $\lambda \in \{0, 0.1, 0.25, 0.4, 0.5, 0.6, 0.75, 1.0\}$ for

Table 2: Hyperparameter search spaces (Optuna, TPE sampler) and selected configurations. The CNN search minimised the validation MSE over up to 150 epochs with median pruning; the Random Forest search maximised the validation R^2 .

<i>CNNs (U-Net and ResUNet++)</i>			
Hyperparameter	Search space	U-Net	ResUNet++
Base width	{32, 48, 64}	48	64
Optimiser	{Adam, AdamW}	Adam	AdamW
Learning rate	$[3 \times 10^{-5}, 5 \times 10^{-4}]$ (log)	3.06×10^{-5}	1.29×10^{-4}
Weight decay	$[5 \times 10^{-6}, 6 \times 10^{-4}]$ (log)	2.15×10^{-5}	1.24×10^{-4}
Dropout	[0, 0.20]	0.001	0.077
Batch size	{8, 16}	16	8
Gradient clipping	{0, 1.0}	1.0	1.0
<i>Random Forest</i>		Selected	
No. of trees	{100, 200, 300, 500}	300	
Max. depth	{10, 15, 20, 30, None}	15	
Min. samples leaf	{1, 5, 10, 20}	1	
Max. features	{sqrt, log2, 0.3, 0.5}	log2	
Min. samples split	{2, 5, 10}	2	

ResUNet++ (eight values $\times$ three seeds); (iii) a leave-one-group-out ablation of the predictor families (six configurations $\times$ three seeds); and (iv) an analysis of the 26-channel meteorological configuration.

375 5. Results

All results are reported on the temporally disjoint 2025 test set, averaged over three seeds (mean $\pm$ standard deviation) unless stated otherwise.

5.1. Model comparison: *RF* vs. *U-Net* vs. *ResUNet++*

Table 3 and Fig. 5 compare the three models on the 22-channel dataset. ResUNet++ trained with the optimal spatial loss ($\lambda = 0.4$, Section 5.2) at-
380 tains the best scores on every metric: $R^2 = 0.208 \pm 0.027$ and SPAEF = 0.297 ± 0.018 . The Random Forest is a solid baseline in terms of R^2 (0.140 ± 0.004) and is remarkably stable, but its spatial pattern fidelity is poor (SPAEF = 0.107): operating per-pixel, it cannot reproduce the coherent drift structures
385 of the snowpack.

Table 3: Model comparison on the 22-channel dataset (2025 test set, mean $\pm$ std over seeds 42/123/7). All CNNs are evaluated at their best-validation epoch (see Section 4.6). Best value per column in bold. SPAEF for RF is computed by reassembling per-pixel predictions into tiles.

Model	$R^2\uparrow$	SPAEF $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	Bias	R^2 std
Random Forest	0.140	0.107	0.515	0.675	-0.307	± 0.004
U-Net (GroupNorm)	0.054	0.186	0.536	0.707	-0.333	± 0.124
ResUNet++ ($\lambda=0.4$)	0.208	0.297	0.477	0.648	-0.282	± 0.027

The plain U-Net (with Group Normalisation, as in ResUNet++, so that the two share the same normalisation; see Section 4.2) is, by contrast, highly unstable across seeds: its R^2 is 0.054 ± 0.124 – a standard deviation more than twice the mean – ranging from -0.064 to 0.184 , the latter approaching
 390 ResUNet++. Its variance is an order of magnitude larger than that of ResUNet++ (± 0.027) and of the RF (± 0.004), so a single run could easily be mistaken for either a failure or a near-state-of-the-art result.

The most instructive comparison is between the U-Net and the RF, whose ranking *reverses* depending on the metric. By R^2 the per-pixel RF (0.140)
 395 clearly leads the U-Net (0.054); by SPAEF the order flips, with the U-Net (0.186) ahead of the RF (0.107). The CNN, even when it lags in pixel-wise accuracy, reproduces the spatial *pattern* of the snowpack better than the RF, which operates per pixel and over-smooths the field. This metric-dependent reversal is precisely the kind of effect that an R^2 -only evaluation would miss,
 400 and it motivates our use of pattern-aware metrics and a multi-seed protocol. ResUNet++ remains the best model on every metric and by far the most reproducible.

Figure 6 illustrates these differences qualitatively on a representative 2025 test tile: both ResUNet++ and the U-Net recover the wind-driven accumu-
 405 lation bands, sharper and closer to the LiDAR reference than the Random Forest, which yields a smoother, washed-out field – consistent with the much lower SPAEF of the per-pixel baseline.

5.2. Spatial-loss weight sweep

Table 5 and Fig. 7 report the effect of the spatial-loss weight λ on ResUNet++. Both R^2 and SPAEF follow a clear concave trend: starting from
 410 pure MSE ($\lambda = 0$), adding a moderate amount of spatial correlation improves performance up to an optimum at $\lambda = 0.4$, after which over-weighting the

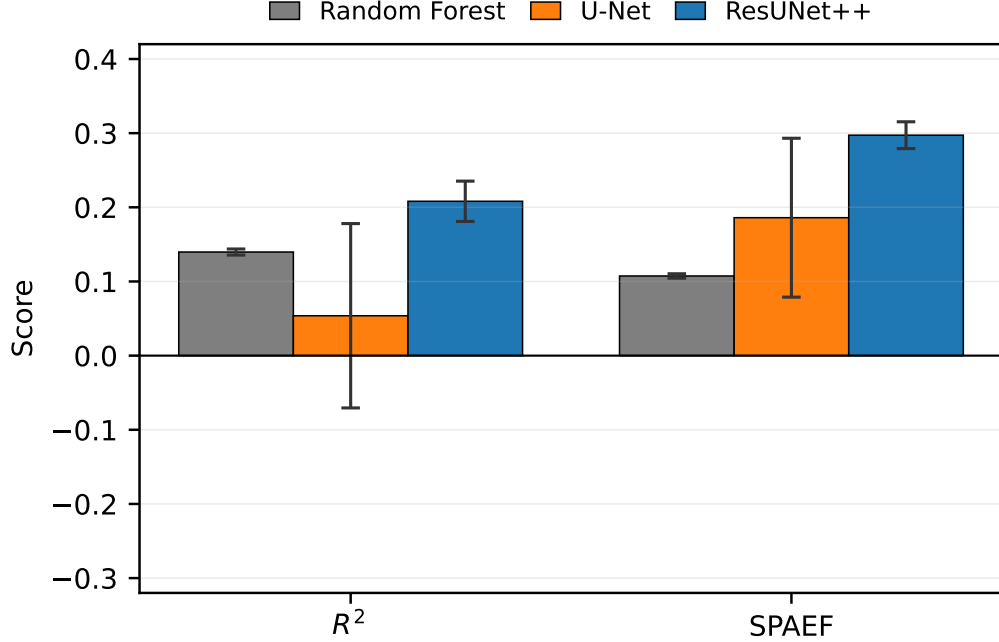


Figure 5: Model comparison on the 2025 test set. Bars show the mean over three seeds and error bars the standard deviation, for R^2 (left) and SPAEF (right). ResUNet++ ($\lambda=0.4$) dominates on both metrics. Note the reversal between the RF and the U-Net: the RF leads on R^2 but the U-Net leads on SPAEF; the U-Net also shows by far the largest seed variance.

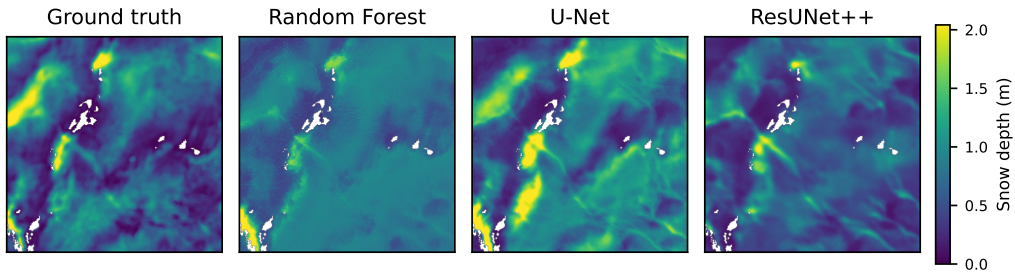


Figure 6: Qualitative comparison of predicted snow-depth maps for a representative 2025 test tile. From left to right: LiDAR ground truth, Random Forest, U-Net (GroupNorm) and ResUNet++ ($\lambda=0.4$). The two CNNs reproduce the wind-driven accumulation bands, whereas the Random Forest produces a smoother, lower-contrast field. White pixels are masked (no valid LiDAR return); all panels share the colour scale.

Table 4: Per-seed R^2 and SPAEF, illustrating the seed-to-seed (in)stability of each model.

Model	R^2			SPAEF		
	s42	s123	s7	s42	s123	s7
Random Forest	0.140	0.136	0.144	0.104	0.110	0.108
U-Net (GroupNorm)	0.042	-0.064	0.184	0.161	0.094	0.304
ResUNet++ ($\lambda=0.4$)	0.186	0.200	0.239	0.294	0.281	0.317

Table 5: Spatial-loss weight sweep for ResUNet++ on the 22-channel dataset (mean $\pm$ std over seeds 42/123/7). The optimum is $\lambda = 0.4$ for both metrics, which also yields the lowest variance.

λ	R^2	R^2 std	SPAEF	SPAEF std
0.00	0.139	± 0.111	0.212	± 0.123
0.10	0.173	± 0.025	0.267	± 0.018
0.25	0.152	± 0.079	0.272	± 0.034
0.40	0.208	± 0.027	0.297	± 0.018
0.50	0.203	± 0.019	0.288	± 0.010
0.60	0.145	± 0.045	0.235	± 0.032
0.75	0.181	± 0.076	0.278	± 0.043
1.00	0.119	± 0.035	0.194	± 0.060

pattern term (towards $\lambda = 1.0$, i.e. pure correlation) degrades both metrics. At the optimum, R^2 rises from 0.139 ± 0.111 ($\lambda = 0$) to 0.208 ± 0.027 , and SPAEF from 0.212 ± 0.123 to 0.297 ± 0.018 – the spatial loss not only improves the mean but also *halves* (in fact reduces by $\sim 80\%$) the seed-to-seed variance, making training markedly more reproducible.

5.3. Channel ablation

To quantify the contribution of each predictor family we perform a leave-one-group-out ablation on ResUNet++, removing one channel family at a time and retraining from scratch (three seeds each). Table 6 and Fig. 8 report the resulting test scores; larger drops relative to the full model indicate more important channels.

Two families are critical. Removing the 5 m (coarse) topography collapses the model to $R^2 = -0.114$ and SPAEF = 0.076, and removing the snow persistence channels is equally damaging ($R^2 = -0.148$, SPAEF = 0.129).

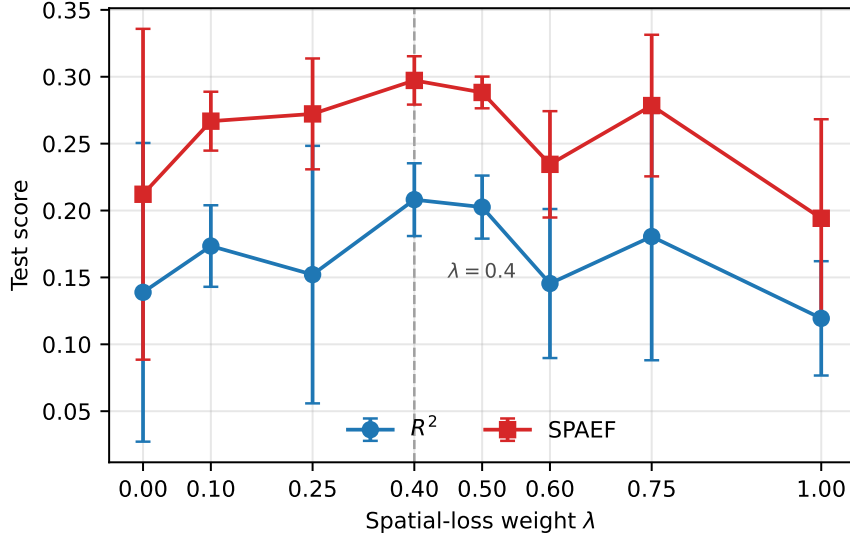


Figure 7: Effect of the spatial-loss weight λ on ResUNet++. Curves show the mean R^2 and SPAEF over three seeds with ± 1 std error bars. Both metrics peak at $\lambda = 0.4$ (dashed line); the error bars also contract around the optimum, indicating improved reproducibility.

Table 6: Leave-one-group-out channel ablation for ResUNet++ (22-channel full model, mean $\pm$ std over three seeds). Rows are sorted by impact; the largest degradations (most important channels) are at the top.

Configuration	R^2	SPAEF
Full (22 channels)	0.132 ± 0.112	0.251 ± 0.051
– Snow persistence	-0.148 ± 0.113	0.129 ± 0.026
– Topography 5 m	-0.114 ± 0.119	0.076 ± 0.080
– S_x wind sheltering	0.089 ± 0.097	0.225 ± 0.044
– Topography 1 m	0.079 ± 0.092	0.220 ± 0.048
– SCE (snow-cover extent)	0.132 ± 0.027	0.242 ± 0.022

This confirms that meso-scale terrain context and recent snow history are the backbone of the prediction. The S_x wind-sheltering channels and the 1m topography are moderately important, while the instantaneous satellite snow-cover extent (SCE) is nearly redundant – its removal leaves R^2 essentially unchanged and even reduces variance, consistent with the persistence channels already encoding the relevant snow-cover information.

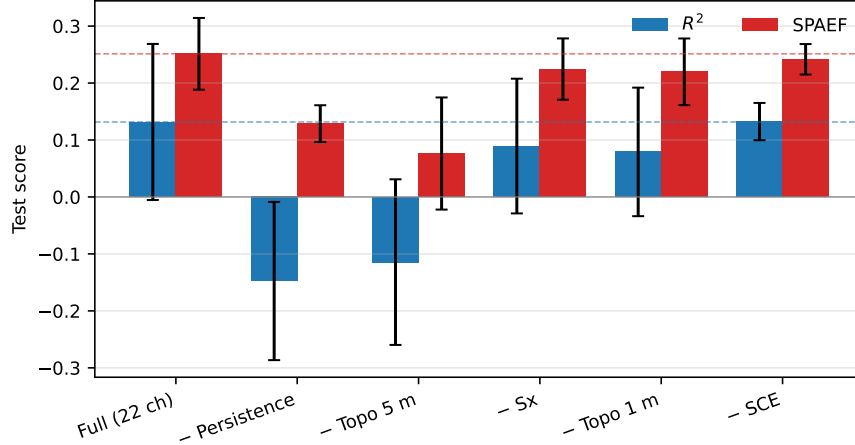


Figure 8: Channel-family ablation for ResUNet++. Bars show the test R^2 and SPAEF obtained when each predictor family is removed (mean $\pm$ std over three seeds); dashed lines mark the full 22-channel model. Removing snow persistence or 5 m topography drives R^2 below zero, identifying them as the most influential predictors, whereas SCE is nearly redundant.

Note. The ablation is performed at $\lambda = 0$ (pure MSE) so that all configurations are compared under identical, loss-neutral conditions; the full-model
 435 baseline in Table 6 therefore corresponds to the $\lambda = 0$ row of Table 5 and is lower than the $\lambda = 0.4$ optimum used elsewhere.

5.4. Meteorological channels (extended configuration)

Finally, we test whether adding explicit meteorological forcing improves prediction. Table 7 compares the best 22-channel ResUNet++ ($\lambda = 0.4$) with
 440 the 26-channel configuration that adds the four temperature/precipitation channels; for the latter the best operating point of the λ sweep was $\lambda = 0.6$. The two configurations are statistically indistinguishable on the central metrics: the 26-channel model is marginally better on R^2 (0.213 vs. 0.208) and the 22-channel model is marginally better on SPAEF (0.297 vs. 0.283), with
 445 overlapping standard deviations on every metric. Moreover, the 22-channel model is *more stable* (R^2 std ± 0.027 vs. ± 0.054). This holds across the whole spatial-loss sweep (Fig. 9): the two configurations track each other within the seed variance at every λ , and the meteo-augmented model is generally the more erratic of the two.

450 We conclude that, for this catchment and predictor set, the topographic

Table 7: Effect of adding meteorological channels. Best 22-channel ResUNet++ ($\lambda=0.4$) vs. best 26-channel configuration ($\lambda=0.6$); mean $\pm$ std over three seeds.

Configuration	R^2	SPAEF	MSPAEF	MAE	RMSE	R^2 std
22-ch ($\lambda=0.4$)	0.208	0.297	0.462	0.477	0.648	± 0.027
26-ch ($\lambda=0.6$)	0.213	0.283	0.463	0.480	0.646	± 0.054

and snow-history channels already capture most of the explainable variance, and the scalar (spatially uniform) meteorological channels add little complementary spatial information. We therefore retain the 22-channel configuration as our reference model and report the meteorological experiment as a complementary analysis.

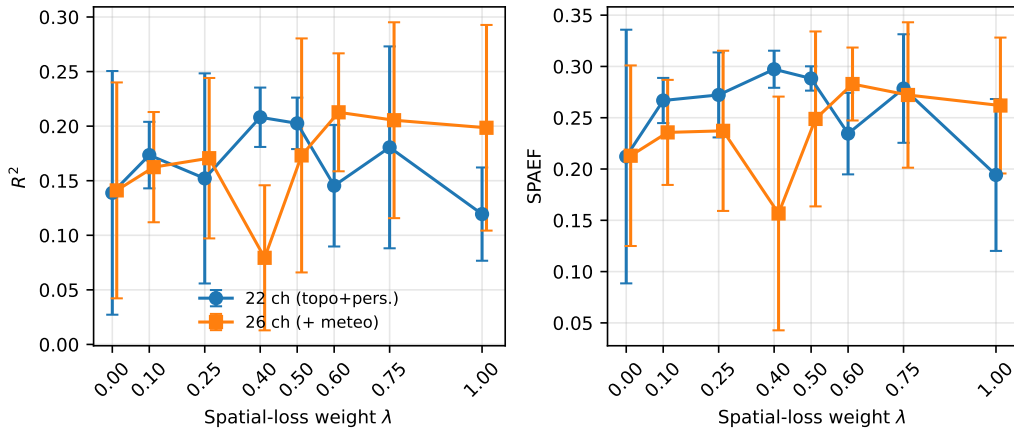


Figure 9: Comparison of the 22-channel and 26-channel (with meteorological forcing) ResUNet++ configurations across the spatial-loss weight λ : R^2 (left) and SPAEF (right), mean $\pm$ std over three seeds. Differences fall within the seed variance at every λ , indicating no significant benefit from the added meteorological channels.

5.5. Spatial vs. temporal generalisation

All results so far use the temporal split, which tests the hardest and most operationally relevant scenario: extrapolation to an unseen – and anomalously snowy – year. To disentangle this temporal difficulty from the model’s ability to generalise in space, we re-evaluate the reference ResUNet++ ($\lambda = 0.4$) on a purely *spatial* split of the same tiles. The catchment is partitioned into three contiguous bands along the easting (Fig. 10); because the 256 m

465 tiles overlap by 50% (a 128 m stride), the tiles that straddle a band boundary are dropped, leaving a ≥ 256 m buffer so that no pixel is shared between splits. Every spatial location is assigned to a single split across all dates, so the model is trained and tested on the same years (2021–2025) but on disjoint *locations*.

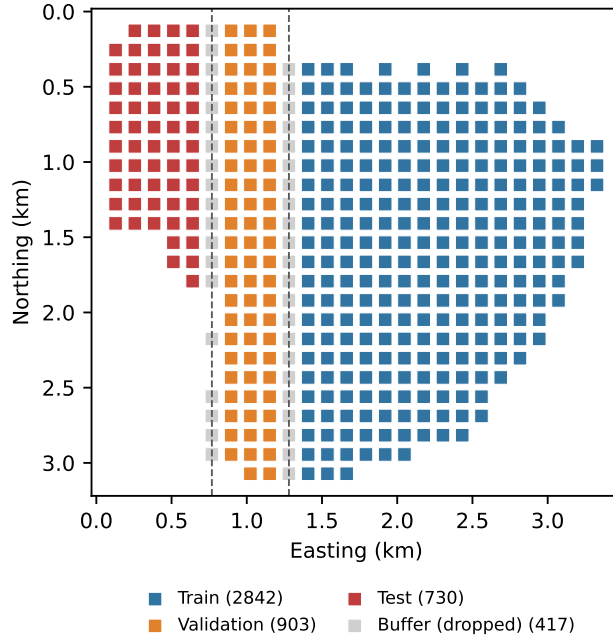


Figure 10: Spatial train/validation/test split of the Izas tiles. Each marker is a tile location; the three contiguous easting bands are separated by a dropped buffer (grey, dashed boundaries) that removes all overlap between splits. Every location keeps the same split across all acquisition dates, so the split is purely spatial.

Table 8 compares the two regimes. Under the spatial split the model is markedly more accurate: R^2 roughly doubles (0.409 ± 0.026 vs. 0.208 ± 0.027), SPAEF rises to 0.448 and MSPAEF to 0.723, and the negative bias shrinks (-0.211 vs. -0.282). The MAE and RMSE are slightly higher because the spatial test set spans every year, including the deep-snow 2025 magnitudes, but the fraction of variance explained is far larger. The seed variance remains small (± 0.026), confirming that the instability seen elsewhere is tied to the temporal extrapolation rather than to the architecture. In short, the model generalises well across *space*; the dominant source of error is the temporal

Table 8: Reference ResUNet++ ($\lambda=0.4$) under the temporal and the spatial split (2025 vs. unseen-locations test set; mean $\pm$ std over seeds 42/123/7). The spatial split, which keeps all years in training, is substantially easier.

Split	R^2	SPAEF	MSPAEF	MAE	RMSE	Bias
Temporal (2025)	0.208 ± 0.027	0.297 ± 0.018	0.462	0.477	0.648	-0.282
Spatial (locations)	0.409 ± 0.026	0.448 ± 0.023	0.723	0.486	0.693	-0.211

extrapolation to an atypical year, which we therefore retain as our primary, more demanding benchmark.

6. Discussion

6.1. Pattern-aware metrics change the ranking

The headline finding of our study is methodological: *how* a high-resolution snow model is evaluated can change which model one would select. Judged by R^2 alone, the per-pixel Random Forest (0.140) is a strong second behind ResUNet++ (0.208), ahead of the U-Net (0.054). Judged by SPAEF, the order of the two baselines *reverses*: the U-Net (0.186) now ranks above the RF (0.107), and the gap between the RF and ResUNet++ widens to nearly a factor of three, revealing that the RF reproduces the spatial pattern far worse than its R^2 suggests. The discrepancy arises because R^2 rewards getting the overall magnitude right and is dominated by the easy, smoothly varying part of the field, whereas SPAEF jointly penalises errors in spatial correlation, dispersion and value distribution – the components that encode the wind-driven drift structures of interest. For applications that depend on the *location* of deep and shallow snow (avalanche release zones, meltwater routing, ecological niches), SPAEF/MSPAEF are the operationally relevant criteria, and we recommend reporting them alongside pixel-wise metrics in future high-resolution snow studies.

6.2. Architecture and spatial loss matter

Both CNNs share the same input stack, training protocol and Group Normalisation, so the contrast between them isolates the effect of the *architecture*. Under these identical conditions the plain U-Net is highly unstable across seeds ($R^2 = 0.054 \pm 0.124$, ranging from -0.064 to 0.184), whereas ResUNet++ is both more accurate and far more reproducible ($R^2 =$

0.208 $\pm$ 0.027). We attribute this to the residual connections and the attention/ASPP blocks of ResUNet++, which provide a well-conditioned optimisation landscape – the network can learn incrementally from an identity mapping and aggregate multi-scale context – both critical with only 27 LiDAR dates, where a plain encoder–decoder is highly sensitive to initialisation. (We also note that Batch Normalisation, our first choice for the U-Net, was unusable here: its running statistics collapsed at evaluation under the small batches, which is why we use Group Normalisation in both networks.)

The spatial loss further improves *reproducibility*: the R^2 standard deviation of ResUNet++ drops from ± 0.112 at $\lambda = 0$ (pure MSE) to ± 0.027 at the optimum $\lambda = 0.4$ (Table 5), a fourfold reduction in seed variance on top of the accuracy gain. In an operational setting where a single model must be trained and trusted, this stability is arguably as valuable as the accuracy itself.

6.3. What drives the prediction?

The ablation identifies snow persistence and 5 m topography as the two indispensable predictor families, with the S_x wind-sheltering indices and 1 m topography as secondary contributors. This is physically coherent: recent snow history is a strong autoregressive prior on current depth, and meso-scale terrain governs the wind redistribution that creates the dominant accumulation patterns at Izas. The near-redundancy of the instantaneous satellite SCE is explained by the persistence channels, which are themselves temporal aggregates of the same snow-cover product and carry the relevant information in a less noisy form.

6.4. Why meteorological forcing did not help

Adding temperature and accumulated-precipitation channels produced no significant improvement. We see two reasons. First, the meteorological inputs are *scalar per date*: they shift the whole tile up or down but provide no spatial structure, so they can only help to the extent that the date-level snow magnitude is not already inferable from the snow-history channels – which it largely is. Second, the small number of distinct acquisition dates limits the amount of temporal signal the network can exploit before over-fitting. A spatially distributed meteorological forcing (e.g. gridded temperature, incoming radiation or modelled wind fields) might carry complementary spatial information; we regard this as a promising direction rather than a negative result about meteorology per se.

6.5. Limitations

Several limitations temper our conclusions. (i) The benchmark is a single, small catchment; transferability to other terrains and snow climates remains to be demonstrated. (ii) The test set is a single, exceptionally snowy year; while this is a deliberate and demanding extrapolation test, it also inflates the seed variance and limits the precision of the absolute scores. The spatial-split experiment (Section 5.5), where R^2 roughly doubles to 0.409, confirms that this temporal extrapolation – rather than spatial generalisation – is the dominant source of error. (iii) The number of LiDAR dates (27) is small by deep-learning standards, which constrains model capacity and amplifies initialisation sensitivity. (iv) The persistence channels treat clouds in the satellite snow-cover product as “no-snow”, which may bias the recent-history estimate during prolonged cloudy periods; a cloud-aware temporal gap-filling could refine this input. (v) MSPAEF is reported only as a complementary, bias-aware indicator; a systematic study of how it ranks models relative to SPAEF is left for future work. Addressing these points – multi-catchment training, more acquisition dates, distributed meteorological forcing and cloud-aware snow-history – defines our future work.

7. Conclusions

We presented a reproducible benchmark for high-resolution (1 m) snow-depth prediction in the Izas experimental catchment of the Central Spanish Pyrenees, built from a 27-date airborne LiDAR time series and a rich stack of topographic, wind-sheltering and snow-history predictors provided in collaboration with IPE-CSIC. Using a strictly temporal train/validation/test split that forces extrapolation to an exceptionally snowy year, we compared a Random Forest baseline with U-Net and ResUNet++ convolutional networks, and we evaluated all models with pattern-aware spatial metrics (SPAEF, MSPAEF) in addition to conventional pixel-wise scores.

Our main conclusions are:

- **ResUNet++ with a spatial loss is the best and most reliable model.** At the optimal spatial-loss weight $\lambda = 0.4$ it achieves $R^2 = 0.208 \pm 0.027$ and $\text{SPAEF} = 0.297 \pm 0.018$, outperforming both the Random Forest ($\text{SPAEF} = 0.107$) and a far less stable plain U-Net ($R^2 = 0.054 \pm 0.124$, $\text{SPAEF} = 0.186$).

- 575
Pattern-aware evaluation is essential. SPAEF reverses the ranking of the Random Forest and the U-Net relative to R^2 , and reveals that the per-pixel RF reproduces the spatial pattern almost three times worse than ResUNet++ – effects that an R^2 -only assessment would miss. We recommend reporting SPAEF/MSPAEF and multi-seed statistics as standard practice in high-resolution snow mapping.
- 580
The hybrid spatial loss helps twice. Blending pixel-wise MSE with a spatial-correlation term improves both accuracy and pattern fidelity and substantially reduces the variance across seeds, with a clear optimum at $\lambda = 0.4$.
- 585
Snow history and meso-scale topography are the key predictors, whereas instantaneous satellite snow-cover extent is nearly redundant, and scalar meteorological forcing does not significantly improve the prediction for this catchment.

By coupling a stable, attention-augmented architecture with a pattern-oriented loss and evaluation, this work provides both a methodological template and a public benchmark for high-resolution snow-depth mapping. Future work will extend the approach to multiple catchments and snow climates, incorporate spatially distributed meteorological forcing and cloud-aware snow-history inputs, and assess transferability across years with contrasting snow conditions.

Code and data availability

The dataset, trained model weights and the code required to reproduce all experiments will be made available in a public repository (Zenodo / GitHub) upon acceptance.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization.
Second Author: Methodology, Software, Validation, Writing – review & editing. **Jesús Revuelto:** Resources, Data curation, Writing – review & editing, Supervision.

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Declaration of competing interest

610 The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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